

Plate&2Recipe

A package for PLATE RECOgnition In PYthon

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1 Introduction

intro (Grady, 2006) Grady (2006) (Grady, 2006)

2 Dependencies

Currently, **Linux is the only supported platform**. Since MacOS is Unix-based, `platerecipy` is expected to be installed and used without a problem. The main reason that Windows is not supported at this point is that there some functionalities are implemented in C and require a slightly different compilation and linking procedure. Until the first release, availability of `platerecipy` on Windows is not a development priority.

Concerning the Python dependencies, **`pip` should automatically install the necessary packages**. Nonetheless, the following includes the necessary dependencies:

- `numpy` (Harris et al., 2020): for array operations
- `scipy` (Virtanen et al., 2020): for specific scientific utilities
- `matplotlib` (Hunter, 2007): for basic plots
- `pyvista` (Sullivan & Kaszynski, 2019): for `vtk` output and 3D visualization
- `scikit-image` (van der Walt et al., 2014): for a Random Walker implementation

In future releases, the Random Walker from `scikit-image` will be replaced by a C implementation tailored for problems that `platerecipy` is designed for. Until then, the `scikit-image` remains as a dependency.

3 Installation

Until the package is released on PyPI, `platerecipe` can be installed using `pip` on the shell as follows:

```
1 cd dist/  
2 python -m pip install platerecipe-??.?.tar.gz
```

where `platerecipe-??.?.tar.gz` is the installation tar file for the desired version. Note that all installation tar files can be found in `dist` subdirectory.

4 Basic interface

`platerecipy` is developed under the notion that it must accommodate users with different dataset formats and from different grid structures (e.g., spherical, YinYang, icosahedral, etc.). Accordingly, as much as possible, the interface must refrain from assuming any format or structure specific layout to the input arguments. Consequently, the trade-off of optimizing for minimal conversion burden on the end user is more processing and higher memory usage. However, this extra cost at the post processing is vanishingly small compared to the original cost of running numerical models.

With the aforementioned principle in mind, all that is expected from the user is to have access to, at least, the following arrays:

- the Cartesian x , y , and z coordinates (will refer to it as `input_xs`, `input_ys`, and `input_zs`),
- as well as the corresponding field values (will refer to it as `input_field`).

Furthermore, though all input arrays are assumed to be `numpy` (Harris et al., 2020) arrays, they can have any shape (flat or multidimensional).

The first step is to generate a grid:

```
1 from platerecipy.grid import SphericalGrid
2
3 # generating a consistent grid for interpolation
4 grid = SphericalGrid(input_xs, input_ys, input_zs)
```

The `SphericalGrid` object is itself a derived type of a generic class `Grid`, but it informs the code whether additional considerations (e.g., applying the wraparound boundary condition or spherical distances) are required. Then, our `SphericalGrid` object is to be used to interpolate `input_field` to the input points to a spherical grid. The segmentation is performed on the linearly-interpolated uniform spherical grid. The interpolation process is performed using the method `interpolate_field`:

```
1 # interpolating an input field
2 field = grid.interpolate_field(input_field)
```

It should be noted that for input fields such as viscosity or strain-rate (which vary by orders of magnitude), the optional argument `take_log` should be set to `True` in order to the linear interpolation is performed in the log space (and then converted back).

With the `grid` object and the interpolated field `field`, we have the necessary arguments to perform plate detection. To initiate the process, a `PlateModel` object is to be constructed as follows:

```
1 from platerecipy.model import PlateModel
2
3 # initializing a plate model
4 m = PlateModel(grid)
```

The newly-created object `m` carries (a pointer) to the `grid` object which includes information about the original input as well as the interpolated grid structure. Though it would have been possible to keep interpolation under the hood (given `PlateModel` carries a `Grid` attribute), the user may want to further manipulate the interpolated field before passing it for plate detection. That is why the two steps are kept separate. The importance of the class `PlateModel` is in the fundamental idea that what defines plate interior/boundary must be made explicit and consistent across subsequent snapshots of the same model. The `PlateModel` class has the ability to stack several fields with

different corresponding weights and normalize the stacked field to span $[0, 1]$. Stacking fields are done through the `PlateModel.stack_field()` method as follows:

```
1 # stacking the interpolated field
2 m.stack_field(field)
```

Once again, for fields that vary by orders of magnitude, `PlateModel.stack_field()` accepts an optional argument `take_log` which if set to `True` it takes the logarithm to flatten the distribution. Most importantly, the stacked fields are assumed to be markers of deformation such that 0 and 1 correspond to minimal and maximal deformation. For a field that is opposite of this (i.e., its high corresponds to low deformation and vice versa), there is an additional argument `invert` which ensures the stacked field stays consistent.

Once all the fields (if multiple) are stacked, the `PlateModel.find_plates()` method performs segmentation:

```
1 # finding plates on the stacked field
2 plate_IDs = m.find_plates(
3     boundary_quantile = 0.9,           # threshold for the boundaries
4     # ...
5 )
```

The `PlateModel.find_plates()` method both returns the segmentation result, and also it save them in the model object as `m.plate_IDs` (containing segmentation IDs) and `m.ID_probs` (containing the associated probabilities).

5 Detailed API documentation

Though this guide is maintained such that with every modification the corresponding section is updated, in case of any discrepancies, function docstrings are to be considered the main source providing updated description. In order to access docstrings, one can easily invoke `help()` command or use `?` on jupyter notebook.

5.1 Overall structure

The source code is effectively divided to two portions:

- `src/platerecipy`: where the Python modules reside, and
- `src/clib`: where various backend utilities are implemented in C for better performance.

5.2 grid module

The `grid` module contains the necessary class and function definitions for grid structure. It contains the generic `Grid` class inherited by `SphericalGrid` class which has the additional attributes relevant for a spherical surface. As a general rule, θ and ϕ correspond to polar (colatitude) and azimuthal (longitude) directions.

```
1 class SphericalGrid(Grid):
2     """
3     Uniform spherical grid.
4     """
5
6     def __init__(
7         self,
8         original_xs : np.ndarray,
9         original_ys : np.ndarray,
10        original_zs : np.ndarray,
11        theta_res   = None,
12        phi_res     = None,
13    ) -> SphericalGrid
14        """
15        Constructs a uniform spherical grid structure.
16
17        If `theta_res` and `phi_res` are provided, the resolution will be
18        determined by those values. Alternatively, the resolution close enough
19        to what provided by the Cartesian arguments.
20
21        Warning
22        -----
23        The Cartesian arguments will be the basis upon which the class method
24        `interpolate_field()` operates. In consistent values will impact the
25        interpolation process.
26
27        Parameters
28        -----
29        original_xs : np.ndarray,
30            array of original input Cartesian x coordinates.
31
32        original_ys : np.ndarray,
```



```

33         array of original input Cartesian y coordinates.
34
35     original_zs : np.ndarray,
36         array of original input Cartesian z coordinates.
37
38     theta_res : int, optional
39         polar (colatitude) resolution.
40
41     phi_res : int, optional
42         azimuthal (longitude) resolution.
43     """
44
45     def interpolate_field(
46         self,
47         field : np.ndarray,
48         take_log = False
49     ) -> np.ndarray:
50         """
51         Interpolates the input field linearly for convex hull interior points and
52         closest points to the exterior.
53
54         Parameters
55         -----
56         field : np.ndarray,
57             array corresponding to the original Cartesian coordinate inputs.
58
59         take_log : bool, False
60             whether to perform the interpolation on the log space (suitable for
61             fields that vary by orders of magnitude), default = False.
62
63         Returns
64         -----
65         np.ndarray
66         """

```

5.3 model module

5.4 io module

5.5 transform module

6 Examples

```
1 from platerecipy.model import PlateModel
2 from platerecipy.grid import SphericalGrid
3
4 # generating a consistent grid for interpolation
5 grid = SphericalGrid(input_xs, input_ys, input_zs)
6
7 # interpolating an input field
8 field = grid.interpolate_field(input_field)
9
10 # initializing a plate model
11 m = PlateModel(grid)
12
13 # stacking the interpolated field
14 m.stack_field(field, take_log=True)
15
16 # finding plates on the stacked field
17 m.find_plates(
18     boundary_quantile = 0.9,          # threshold for the boundaries
19     # ...
20 )
21
22 # outputting as a ParaView readable .vtp file
23 from platerecipy import io
24 io.save_as_vtp(m)
```

References

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A About the backend